

Review retrospective:
*the Gibbs foundation — the definitions every laplace file rests
on, reviewed clean*

Claude Opus 4.8 (1M ctx), reviewer GPT-5.5 Pro

2026-06-05

Abstract

A soundness review of `Laplace/Gibbs.lean`: the seabed’s foundational definitions — `partitionFunction`, `gibbsExpectation`, `gibbsCov` — and the bilinearity / scalar-pull / constant-collapse algebra that every downstream file uses to manipulate Gibbs expectations without unfolding. Result: a **clean fidelity pass with two edits** (a dead **open** and one over-specified header sentence). For a file this central, the value of the review is the confirmation itself: the objects the whole programme is built on say exactly what the primer means.

1 Setting

This is the root of the laplace dependency graph: $Z(t)$, $\langle \cdot \rangle_t$, and $\text{Cov}_t[\cdot, \cdot]$ (with prior $\pi \equiv 1$), plus the algebraic lemmas (`gibbsExpectation_{const,smul,zero,add}`, `gibbsCov_{symm,smul,const,add,zero}` in both slots) that the affine- and multilinear-observable covariance results across the seabed call. Every other laplace file imports it. It had never been independently reviewed.

2 What was reviewed

The three definitions against the primer’s $Z/\langle \cdot \rangle/\text{Cov}$ with $\pi \equiv 1$; and each algebraic lemma’s *hypotheses* — the crux for a definitions file, since the interest is precisely which facts need $Z(t) \neq 0$, which need integrability, and which are unconditional.

3 Fidelity / soundness findings

Sound; no defect. The reviewer confirmed all three definitions faithful (including the implicit Lebesgue measure realising $\pi \equiv 1$) and *no sign error* in the covariance $\langle \varphi \psi \rangle - \langle \varphi \rangle \langle \psi \rangle$. The hypothesis placement is exactly right: `gibbsExpectation_const` genuinely needs $Z \neq 0$ (else Lean’s totalised $/0 = 0$ gives $\langle c \rangle = 0 \neq c$); the `_smul/_zero` and all the `gibbsCov` constant/scalar/zero lemmas are correctly *unconditional*, their docstrings correctly noting that at $Z = 0$ both sides collapse to 0; and the additivity lemmas carry exactly the integrability hypotheses `integral_add` requires, with no spurious $Z \neq 0$. The lone textual mismatch: the module header described the setting as a “smooth potential L ” with “ $t > 0$,” while the definitions and lemmas are stated for arbitrary $L : \mathbb{R} \rightarrow \mathbb{R}$ and arbitrary $t : \mathbb{R}$ — the code is strictly more general than its own prose.

4 Simplifications and documentation fixes applied

One dead open, one prose correction. `open Real` carried nothing (every `Real.exp` is qualified; the only bare `exp` tokens are in the docstring); `open MeasureTheory` stays (bare `Integrable`, `integral_const_mul`, `integral_add`). And the header was relaxed to state that smoothness of L and positivity of t belong to the primer’s setting, not to the generality of the file’s definitions. Build green, no warnings, signatures byte-identical; the reviewer affirmed both. No dead locals, no unnecessary hypotheses, no unused imports were found — a tight foundational file.

5 Disagreements and open questions

None. (One terseness note left report-only: `gibbsCov_add_right`’s docstring, unlike `_add_left`’s, does not restate the integrability hypotheses — accurate as written, since the hypotheses are in the signature.)

6 Lessons

- **For a definitions file, the review *is* the hypothesis audit.** There is no asymptotic constant to re-derive; the entire fidelity question is “does each lemma demand exactly the hypotheses it needs — $Z \neq 0$ where a division must be inverted, integrability where `integral_add` fires, nothing where the totalised $/0 = 0$ makes the statement unconditional — and does the docstring’s $Z = 0$ reasoning hold up?” Confirming that for the root objects is high-value precisely because everything downstream inherits it.
- **Header prose over-specifies more often than it under-specifies, on foundational files.** A definitions file is written to serve a motivating application (here: smooth L , $t > 0$), and the header tends to inherit that framing even though the Lean objects are deliberately general. Worth a glance: do the stated ambient hypotheses actually appear in any signature?

7 Follow-ups

- Remaining never-reviewed `laplace` files are the large `Multi/` analytic files (`RescaledIntegrals` 1547L, and the `Covariance/CovarianceSharp/ CovarianceExplicit` trio, up to 19.6k lines — candidates for sectioned reviews), plus the small root files `AffineKappa3`, `AffineBilinearCov`, `ScalarBound`.